

Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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SCAN: DEMO VIDEO
PAPER + ARXIV

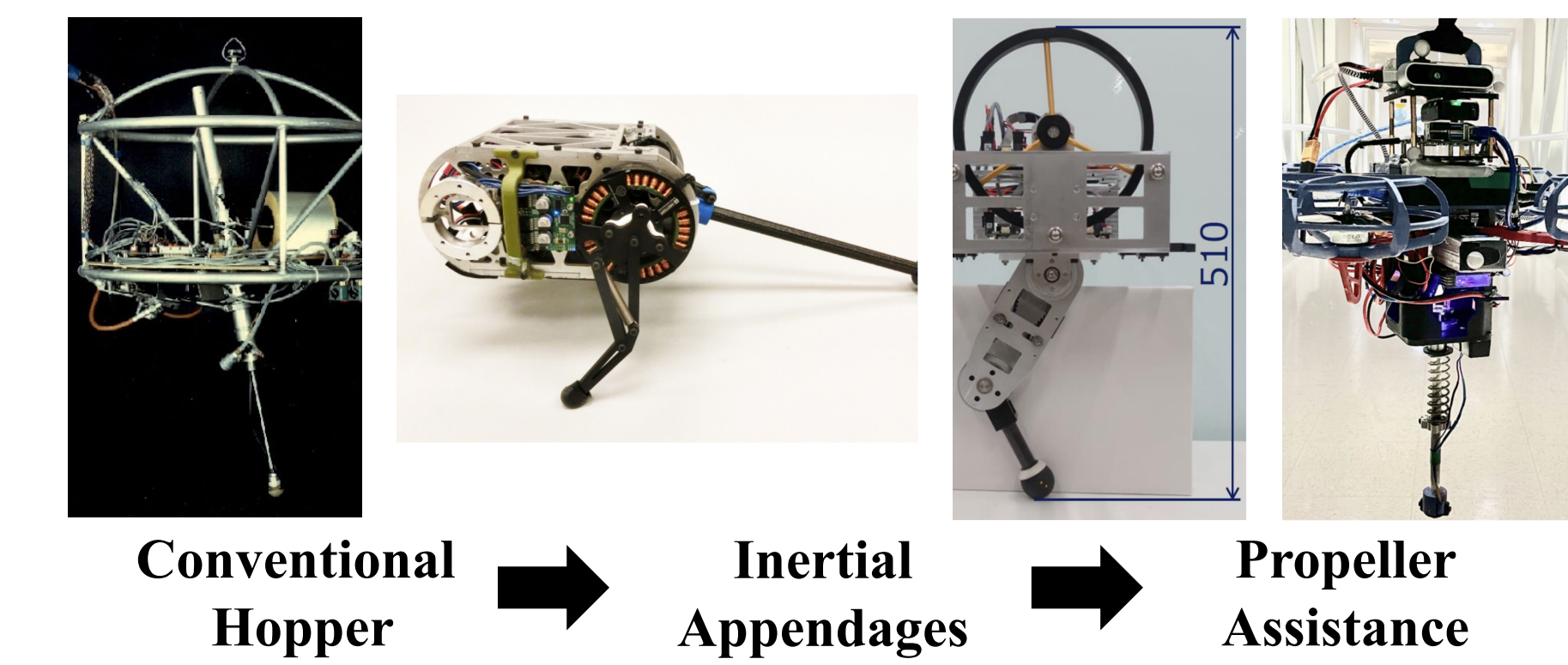


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1 Problem- hybrid dynamics create an authority gap



■ **Stance**
Ground reaction force exists only during a short contact interval.

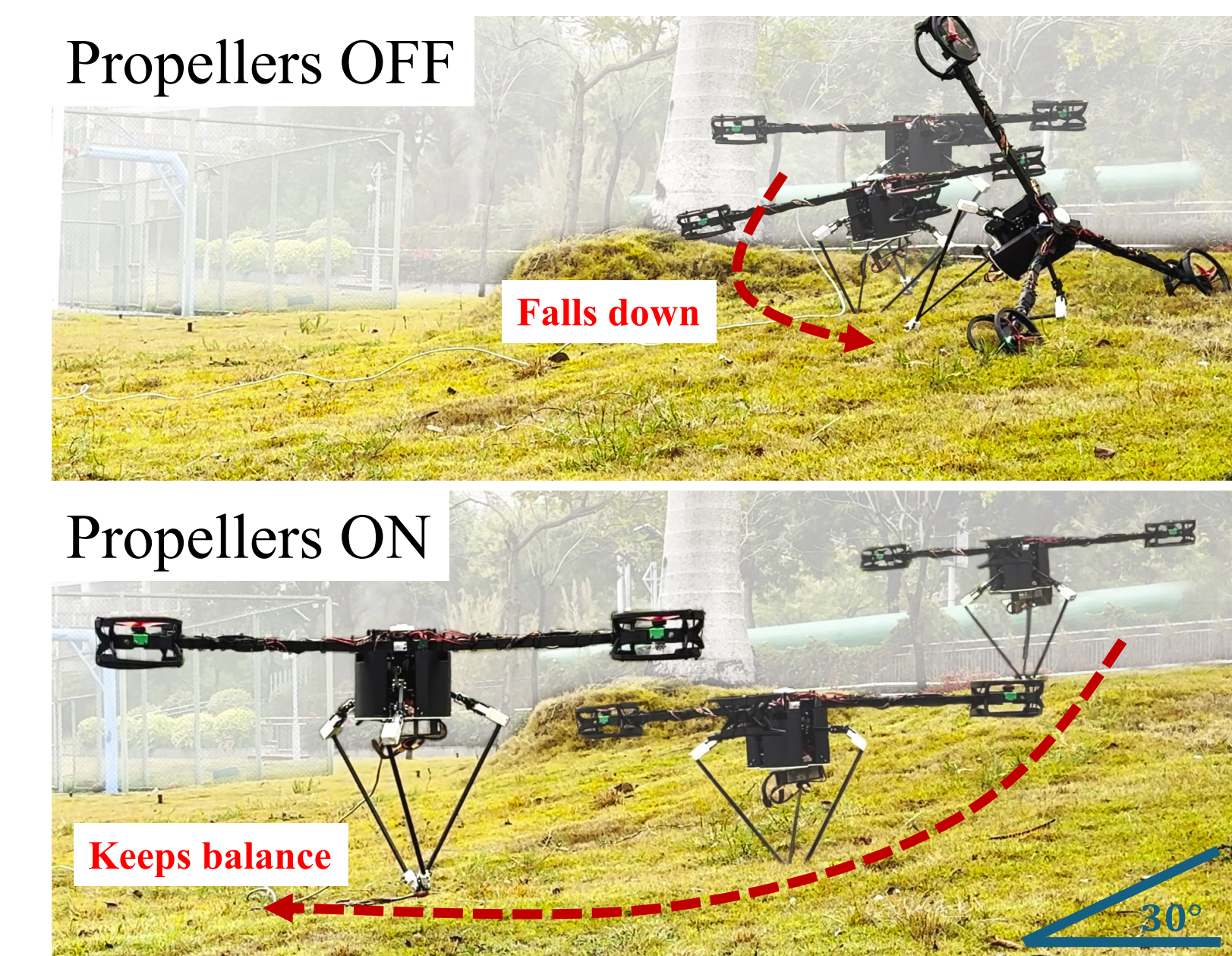
■ **Flight**
No contact: attitude becomes underactuated.

■ **Disturbance**
Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.

Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2 First proof- stable on 30° grass



■ **Propellers off**
After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.

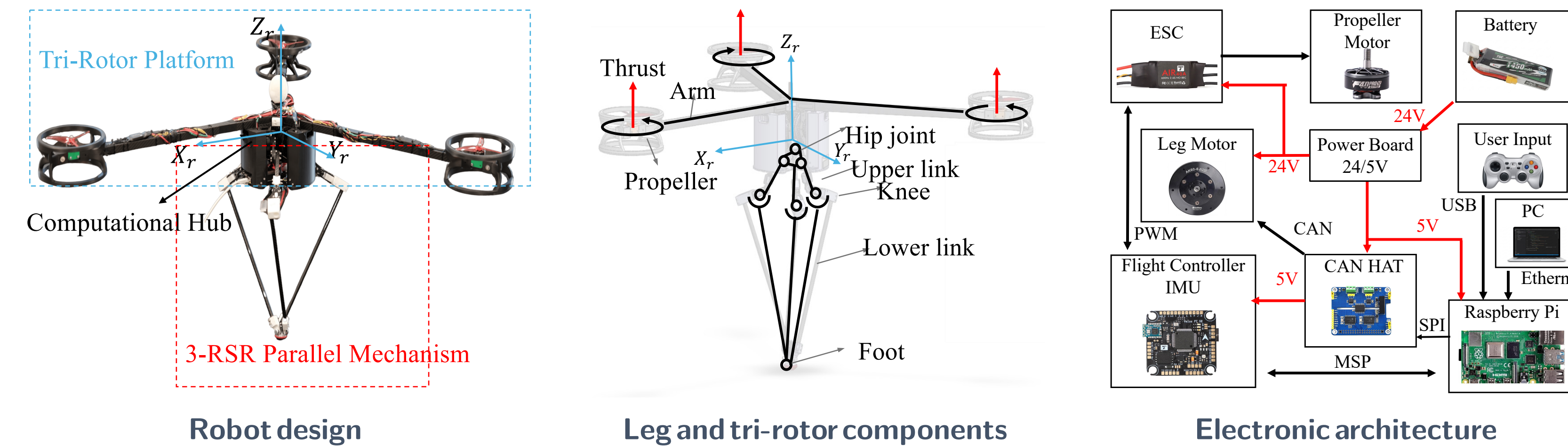
■ **Propellers on**
Differential thrust remains available through flight and restores attitude before the next landing.

■ **What this establishes**
Residual authority in stance.
Primary attitude authority in flight.
Faster recovery from uneven impacts.

Not primary lift: the rotors close the attitude-authority gap.

Propellers off: attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

3 Platform- hardware built for leg-first hopping



■ **3-RSR parallel leg**
Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.

■ **Computational hub**
Batteries and electronics near the hip: low center of mass, low inertia.

■ **Y-layout tri-rotor**
Continuous attitude authority; **10 N** per rotor, total 0.82 mg: a hopper, not a drone.

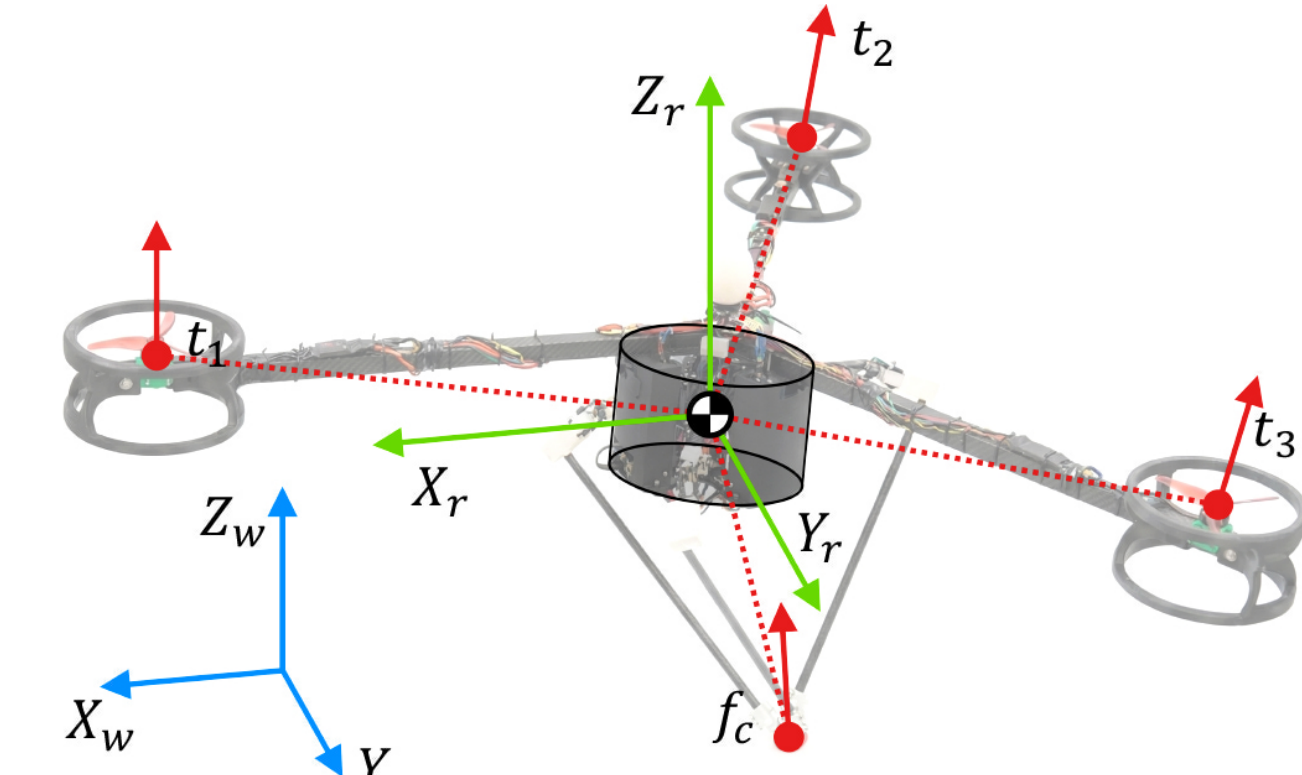
■ **External PC @ 500 Hz**
State machine, estimation, wrench planning, and the HFA allocator.

■ **Raspberry Pi interface**
CAN to the three leg motors; MSP to the flight controller.

■ **Real-time link**
LCM over Ethernet: RTT ≈ 0.2 ms, 99% < 0.5 ms, within the 2 ms period.

3.75 kg total ■ 9.2% leg mass ■ 20 Nm / leg motor ■ 500 Hz control

4 Model- one rigid body, two actuator sets



■ **Net wrench = leg contact + three rotors**

$$\begin{bmatrix} \mathbf{F}_{net} \\ \boldsymbol{\tau}_{net} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_c \\ \mathbf{r}_c \times \mathbf{f}_c \end{bmatrix} + \sum_{i=1}^3 \begin{bmatrix} \mathbf{d}_i \\ \mathbf{r}_i \times \mathbf{d}_i \end{bmatrix} t_i$$

■ **Newton-Euler dynamics of the CoM**

$$m\mathbf{a} = \mathbf{F}_{net} + m\mathbf{g}, \quad \dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{R}^\top \boldsymbol{\tau}_{net} - \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega})$$

Contact force $\mathbf{f}_c$ and thrusts $t_{1,2,3}$ produce the body wrench.

■ **Symbols (values used in all experiments)**

$\mathbf{f}_c$: ground reaction force (world)

$\mathbf{r}_c, \mathbf{r}_i$: arms: CoM → contact, rotor i

$t_i \in [0, 10]$ N thrust along $\mathbf{d}_i = \mathbf{R}\mathbf{e}_3$

$\mathbf{F}_{net}, \boldsymbol{\tau}_{net}$: net force, moment

$m=3.75$ kg; $\mathbf{g}$: gravity; $\mathbf{I}$: inertia

$\mathbf{R} \in \text{SO}(3)$ attitude; $\boldsymbol{\omega}$: body rate

$\mathbf{J}$: leg Jacobian; $\tau_{max}=20$ Nm limit

$\mu=0.4$ friction coefficient

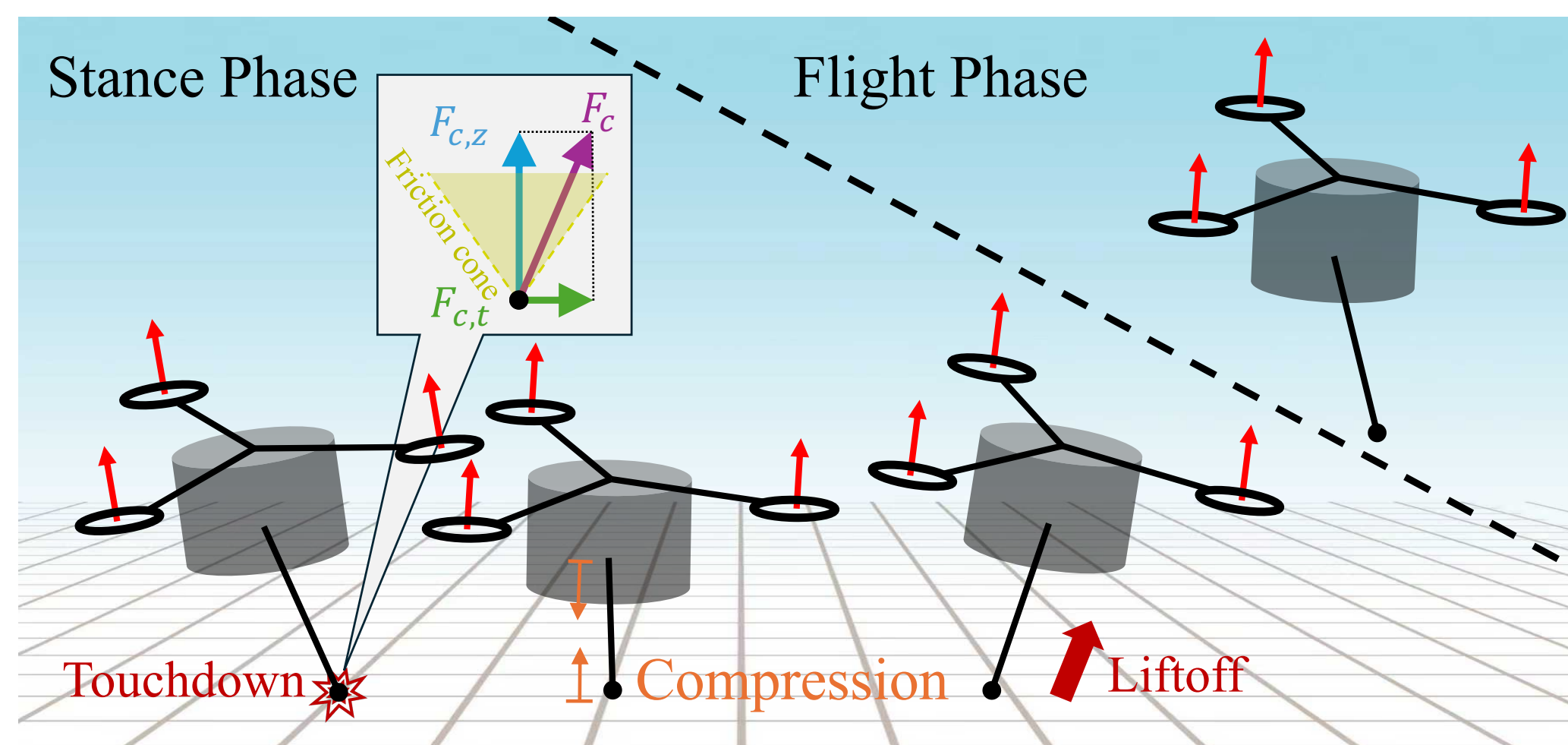
$p_z, \dot{p}_z$: CoM height, velocity; $h_{des}=0.664$ m

l : leg length; $l_0=0.464$ m; Δ_{td} : threshold

k_z, b_z, k_E : vertical, k_v, k_r : Raibert gains

$\mathbf{K}_R, \mathbf{K}_\omega$: attitude PD gains

5 Onehop- event-driven authority handoff



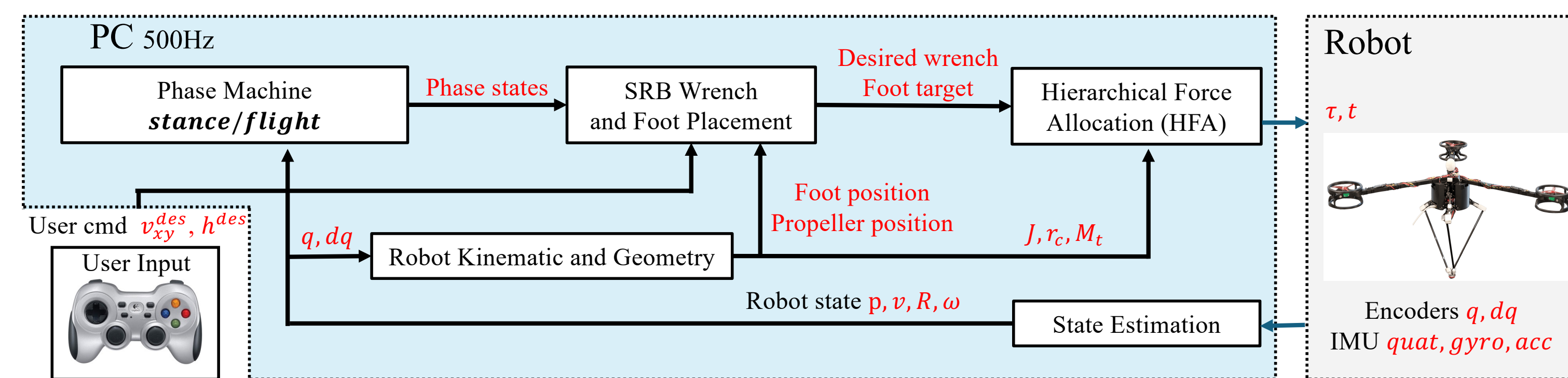
■ **Touchdown:** $l \leq l_0 - \Delta_{td}$
Leg absorbs impact; energy law drives the next apex.

■ **Lift-off:** $l > l_0$
Contact wrench vanishes; rotors primary for attitude.

■ **Handoff**
Events switch authority; no hardware or gain change.

Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

6 Control- leg-priority hierarchical force allocation



500 Hz architecture: commands + feedback → planner → two-stage allocator → leg torques + rotor thrusts.

■ **A. Vertical energy hopping (stance)**

$$E_{sys} = \frac{1}{2} m \dot{p}_z^2 + m g p_z, \quad E_{des} = m g h_{des}$$

$$\mathbf{f}_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z \dot{p}_z + k_E(E_{des} - E_{sys}))$$

Injects push-off energy, absorbs impact.

■ **B. SO(3) attitude PD (both phases)**

$$\mathbf{e}_R = \frac{1}{2} (\mathbf{R}_{des}^\top \mathbf{R} - \mathbf{R}^\top \mathbf{R}_{des})^\vee$$

$$\boldsymbol{\tau}_b^{des} = -\mathbf{K}_R \mathbf{e}_R + \mathbf{K}_\omega (\boldsymbol{\omega}^{des} - \boldsymbol{\omega})$$

$$\text{Roll/pitch only; yaw left free. } \boldsymbol{\tau}^{des} = \mathbf{R} \boldsymbol{\tau}_b^{des}.$$

■ **C. Leg first - constrained least squares**

$$\mathbf{f}_{c,xy}^{cmd} = \arg \min_{\mathbf{f}_{xy}} \|\mathbf{r}_c \times \mathbf{f}_c - (\boldsymbol{\tau}^{des})_{rp}\|_2^2$$

$$\text{s.t. } \|\mathbf{f}_{xy}\|_2 \leq \mu \mathbf{f}_{c,z}^{des}, \quad \|\mathbf{J}^\top (-\mathbf{R}^\top \mathbf{f}_c)\|_\infty \leq \tau_{max}$$

The leg realizes all it can under friction and torque limits.

■ **D. Rotors take only the residual**

$$\boldsymbol{\tau}_{res}^{cmd} = \boldsymbol{\tau}^{des} - \mathbf{r}_c \times \mathbf{f}_c^{cmd}$$

$$\delta \mathbf{t}^{cmd} = \arg \min_{\delta \mathbf{t}} \|\mathbf{M}_t \delta \mathbf{t} - (\boldsymbol{\tau}_{res}^{cmd})_{rp}\|_2^2$$

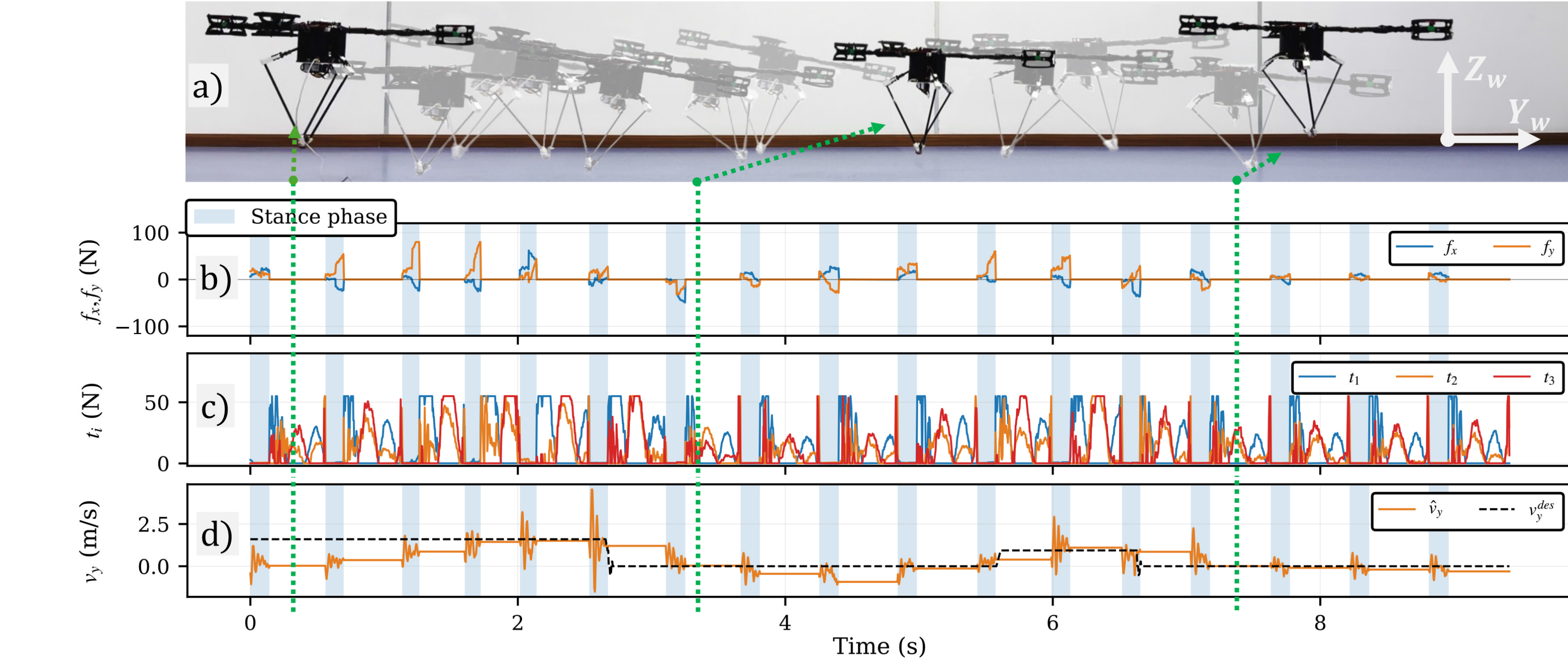
$$\text{s.t. thrust bounds; } \mathbf{t}^{cmd} = \frac{T_{base}}{3} \mathbf{1}_3 + \delta \mathbf{t}^{cmd}, T_{base} = 0.02 \text{ mg.}$$

■ **E. Flight - rotors primary for attitude; Raibert-style foot placement**

$$\mathbf{f}_c = 0; \quad \text{rotor track } (\boldsymbol{\tau}^{des})_{rp}; \quad \mathbf{r}_{td,xy}^{des} = k_v \mathbf{v}_{xy} - k_r \mathbf{v}_{xy}^{des} \Rightarrow \mathbf{p}_{foot}^{des} = [x_{td}^{des}, y_{td}^{des}, -\sqrt{l_0^2 - \|\mathbf{r}_{td,xy}^{des}\|_2^2}]^\top$$

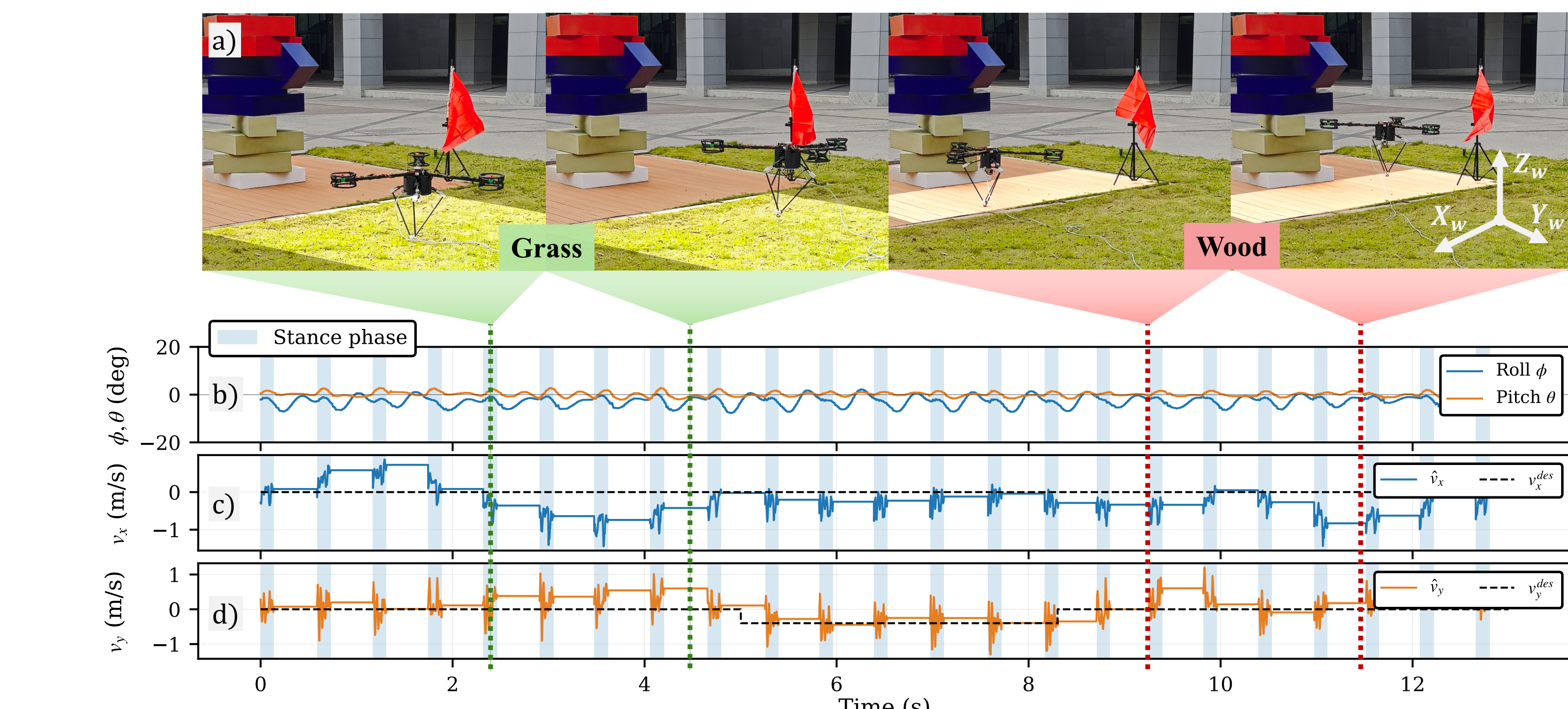
By construction the rotors can only add to - never replace - the leg's contact action.

7 Validation- indoor stop-and-go tracking



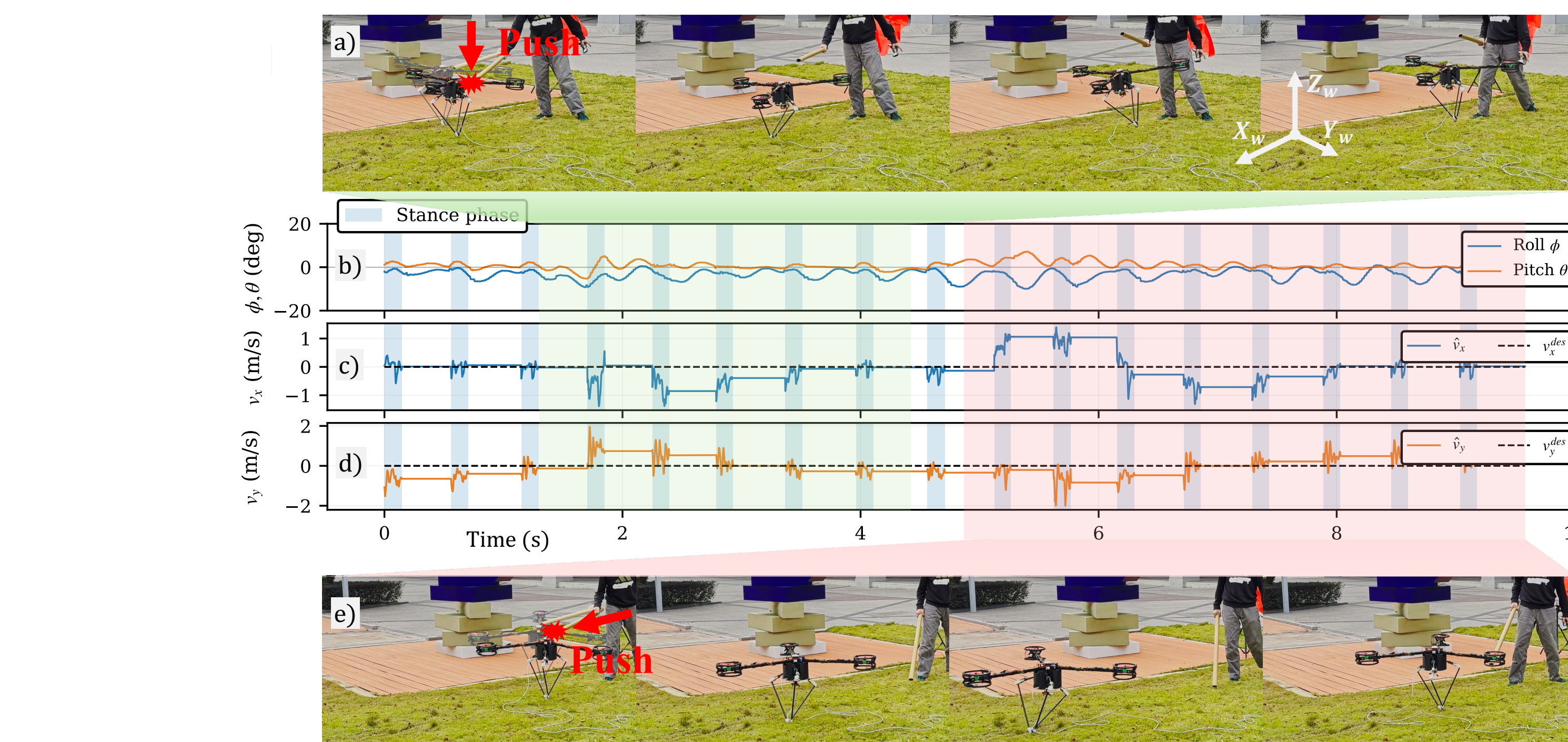
Stop-and-go commands: accelerate, then hard stop (**peaks 1.6 and 0.94 m/s**) - tracked stride by stride. **17 hops** - attitude bounded through every impact: roll $[-15.8^\circ, 3.1^\circ]$, pitch $[-3.3^\circ, 5.6^\circ]$.

8 Validation- grass to wood



Grass → wood under light wind: **23 hops** - same controller, no terrain model, no gain retuning. Attitude stays tight: roll $\sigma = 2.1^\circ$, pitch $\sigma = 1.0^\circ$; deviations decay within 1-2 hops.

9 Validation- push recovery



Rotor-arm push: propellers restore roll/pitch in flight - stable again in **1-2 hops**. Body push (≈ 1.4 m/s velocity spike): foot placement recovers tracking in **3 hops**.